Sri Karthik Chigurupati

Embedded Systems Engineer · PCB Design, Low-Power Firmware and Wireless

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Professional Summary

Embedded systems engineer who takes products from schematic to tested hardware: multilayer PCB design, low-power firmware on nRF52, ESP32-S3 and STM32, and BLE/NFC links to iOS and Android apps. Experience spans medical sensing, cold-plasma equipment, home-safety devices, FPGA signal processing and EV telemetry.

- **Production hardware:** designed a production-ready, low-power 6-layer HDI board and managed component sourcing and functional testing of production PCBs.
- **Biomedical sensing:** validated ECG, SpO2, temperature and BLE on an nRF52840-based health-monitoring prototype and worked on its LiPo BMS circuit design.
- **Ownership:** built InsulinAI, a health-tech project that won 1st place at Emerge21 2025, and led a 20-member electronics team that won business-plan and cost-optimization awards.

Technical Skills

Firmware: Embedded C, FreeRTOS, TouchGFX, MicroPython, Python, Verilog

MCUs and SoCs: nRF52 (nRF52840), ESP32-S3, ESP32, ESP8266, STM32 (F4, G0, H7), NXP S32K, Raspberry Pi, Arduino Opta PLC; FPGA: Artix-7 (Cmod A7), Zynq-7000, Altera MAX 10

Hardware design: Altium, KiCad, LTspice, RapidHarness; 6-layer HDI layout, impedance-controlled RF routing, ultra-low-power sleep architecture, LiPo BMS circuits, Li-ion pack design

Wireless and buses: BLE, NFC, LoRa, XBee, CAN, I2C, SPI, UART

Signal processing: MATLAB (FFT power spectra, spectrograms, matched filtering), fixed-point arithmetic

Test and bring-up: biomedical sensor validation (ECG, SpO2, temperature), functional PCB testing, evaluation boards and lab instruments

Tools: STM32CubeIDE, Keil uVision, MPLAB IDE, Mbed Studio, Arduino IDE / PLC IDE, Android Studio, Xilinx Vivado, Vitis HLS, ModelSim, 4D Systems

Experience

Electronics Engineer — Augeas Technologies

Hyderabad, TG · Dec 2025 – Present

- Designed and routed a production-ready 6-layer HDI PCB for an ESP32-S3 Edge AI monitoring device, with impedance-controlled RF switching and an ultra-low-power hardware sleep architecture.
- Contributed to the NFCFind ecosystem, integrating NFC tag hardware with iOS and Android apps for secure asset tracking and lending workflows.
- Built the hardware and firmware for Stovyn, a privacy-first infrared stove monitor that warns about heat and unattended cooking without streaming video.
- Led hardware and firmware design of the Plasma Knife Standalone module, integrating a stepper motor, servo drive and Arduino Opta PLC for precise actuation.
- Managed component sourcing and full functional testing of PCBs for precision cold-plasma modules (Plasma Tip Chargers, Plasma Knives) built with partner IonField Systems.

Digital Design Intern — STMicroelectronics

Greater Noida, UP · May 2024 – Jul 2024

- Built an NFC transmitter on an Artix-7 FPGA in Verilog using ASK modulation and Modified Miller encoding, extending communication range up to 1 m.
- Validated it against an STM32 G0B1RE with an X-NUCLEO-NFC06A1 board, and used an Android phone as the receiver to prove end-to-end NFC communication.

Project Intern — Sensate Interface Devices

Hyderabad, TG · Jun 2023 – Jul 2023

- Tested and validated ECG, SpO2 and temperature sensors and BLE communication on the prototypes using evaluation boards and lab instruments, identifying and resolving compatibility issues.
- Debugged prototype firmware on the nRF52840 SoC using the nRF52 DK.
- Worked on the design of the LiPo battery management (BMS) circuit for the prototypes.
- Tested and analyzed power electronics components with cross-functional teams.

Projects and Leadership

Electronics Head and Data Acquisition Engineer — SolarMobil Manipal Manipal, KA · Aug 2022 – Sep 2024

- Led a 20-member team designing embedded and powertrain systems for the solar electric vehicle Zenith (ESVC'23, SEVC'23, SEVC'24).
- Designed a new data acquisition and telemetry system for the BWSC'25 car Heliark on an STM32H750B-DK (Arm Cortex-M7) with industrial-grade sensors, XBee radios and an InfluxDB backend, improving data collection accuracy by 20%.
- Co-developed Heliark's driver display on the same board using TouchGFX (Model-View-Presenter) and FreeRTOS tasks scheduled from timing analysis, showing battery state of charge, temperature, pack voltage, motor current and speed.
- Built Zenith's data acquisition system on STM32 MCUs over CAN, reaching 99.9% uptime at SEVC and ESVC 2023, with real-time analysis tools that guided driving strategy to a 50 km/h average speed.
- Developed a live driver dashboard (STM32F401RE, Raspberry Pi, TFT) and LoRa/ESP8266 telemetry to Firebase for the support vehicle.
- Designed and tested a 72 V, 96 Ah Li-ion battery pack and HV wiring harness (Orion BMS, Kelly KLS motor controller, MPPT, safety-critical systems), and wrote BMS and MPPT configuration software that improved energy efficiency by 15%.
- Managed 300 participants and 150 volunteers at SEVC'24 and served as Technical Judge for electrical inspections.

LFM Signal Generator — Final-Year Thesis Verilog, Vivado, MATLAB · Mar 2025 – Aug 2025

- Designed and verified a DDS-based linear frequency modulated signal generator on a Xilinx Zynq-7000, with precise frequency-sweep control and efficient resource use.
- Built MATLAB signal-processing workflows: FFT power-spectrum analysis, spectrograms, matched filtering and chirp pulse compression.

InsulinAI — Health-Tech Project 2025

- Developed InsulinAI, an insulin-dosage recommendation tool, which won 1st place in the Professionals category at Emerge21 (InsideIIM × Emergent), Nov 2025.

RGB to Halftone Image Converter — AISSM 2024 Final Project, NCKU Taiwan Jul 2024

- Designed a Verilog image-processing pipeline with FSM-based control that converts 24-bit RGB images in ROM to 8-bit grayscale, then applies Floyd–Steinberg error diffusion to produce binary halftone images stored in RAM.
- Applied fixed-point arithmetic with configurable rounding; simulated in Synopsys VCS, synthesized with Design Compiler, and verified outputs against golden reference patterns.

Automated IoT Biofloc System — Research Project ESP32-S3, Blynk IoT · Jan 2024 – Dec 2024

- Built an ESP32-S3 system for real-time aquaculture water-quality control, integrating TDS, pH and dissolved-oxygen sensors with automated pumps, Peltier cooling, a servo fish feeder and a Blynk IoT dashboard.

Education

B.Tech, Electronics and Communication Engineering, Minor in Embedded Systems Oct 2021 – Aug 2025
Manipal Institute of Technology (MAHE), Manipal, KA · Electives: Embedded System Design, Real-Time Systems, FPGA-Based System Design, Internet of Things

AISSM 2024 Summer School — National Cheng Kung University, Tainan, Taiwan Jul 2024 – Aug 2024
Digital IC design and synthesis; semiconductor technology and supply chains

Foundational Programme in Data Science and Programming — IIT Madras (remote) Jan 2022 – Dec 2022

Awards

- **NCKU AISSM 2024 scholarship:** the only bachelor's student from India nominated; awarded a 50% scholarship (USD 1,200).
- **SolarMobil team awards:** Future Award for best design and innovation, and Best Business Plan and Cost Optimisation Award (ESVC 2023); Solar Endurance Award and Optimum Cost Award (SEVC 2023).
- **TechTatva'23:** 4th place in Vedanth 13.0, the national technical exhibition at MIT Manipal, showcasing Zenith's data acquisition and telemetry system.
- **TechTatva'22:** 1st place in Electrobuzz, building a metal-detector circuit to locate hidden targets in a field.

Certifications and Languages

ARM Cortex-M Overview, Introduction to Programming with MATLAB, Essentials of Entrepreneurship: Thinking and Action (all Coursera) · **Languages:** English (IELTS Band 7), Telugu, Hindi